# Luis A. Rodriguez Condit

## Software Architect – Embedded & Web

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## Professional Summary

- Software architect with 20+ years designing and building systems spanning embedded hardware, web platforms, and desktop applications
- Expertise in bridging low-level systems (hardware emulation, C/native code) with modern web architectures (Angular/TypeScript)
- Technical leadership: architect full-stack solutions, lead cross-functional teams, mentor engineers, set technical direction
- Core strengths: systems-level design, WebAssembly integration, performance optimization, Linux/cross-platform development
- Track record of independent delivery and rapid prototyping, including production platforms used by millions of users
- Elected to Texas Instruments' Technical Ladder for technical excellence

## Skills

- **Languages:** JavaScript, TypeScript, C, Java, C++, Python, z80 Assembly
- **Architecture & Systems:** Hardware emulation (Z80, eZ80, T4, Lapis MCU), systems-level design, cross-platform development, Linux, communication protocols
- **Frontend & Web:** SPA architecture (Angular), responsive web design, cross-browser development, WebUSB/WebSerial, accessibility (WCAG), Google Analytics
- **Backend & APIs:** Java Spring Boot, Node.js, REST APIs, OpenAPI/Swagger
- **WebAssembly & Native Integration:** Emscripten, JavaScript/C glue layers, cross-compilation, JNI, WebAssembly
- **DevOps & Build Systems:** Jenkins, Gulp, Ant, CMake, build automation, CI/CD, Git
- **Testing & Quality:** Unit testing (Mocha/Chai/Jasmine/Jest), Selenium WebDriver, test-driven development, code review, linting, sanitizers, static analysis
- **Leadership:** Technical architecture decisions, team leadership, Agile/Scrum, mentorship, cross-functional collaboration

## Professional Experience

### Texas Instruments -- Software Architect -- *(Nov 2010 - Sep 2025, Dallas, TX)*

Built desktop and web applications bridging low-level hardware emulation and proprietary communication protocols, bringing embedded calculator systems to millions of students and educators. Led technical planning and project execution for multiple development teams. Recognized with election to TI's Technical Ladder for technical excellence.

#### Hardware Integration & Cross-Platform Development

- Implemented calculator emulators for Z80, eZ80, Toshiba T4, and Lapis MCU in JavaScript; designed JavaScript/C integration for cross-compiling embedded C libraries to WebAssembly with Emscripten.
- Ported proprietary CARS communication protocol to generate platform-native libraries (Windows/macOS) via JNI for JavaFX desktop applications, establishing unified USB communication behavior across platforms.
- Implemented WebUSB and WebSerial APIs for direct browser-to-hardware communication, enabling web applications to connect to physical calculator devices without drivers or plugins.
- Built ElectronJS desktop application for TI-83 Plus CE emulator, deployed for educators completing France's CAPES certification.

#### Frontend & Web Development

- Led development of multiple SPAs with TypeScript and Angular serving millions of users: calculator emulators, license activation systems, and hardware connectivity tools.

- Designed and implemented license activation system with Angular frontend, Java/Spring Boot middleware, and Oracle backend; integrated OAuth authentication and third-party license validation.
- Established component-based design patterns and responsive layouts across calculator applications, ensuring cross-platform compatibility and accessibility user experiences.
- Implemented screen reader support across multiple calculator platforms using Z80 assembly, JavaScript, and OCR interpretation, enabling WCAG-compliant interaction for vision-impaired users.
- Owned full development lifecycle from requirements gathering through deployment: translated business needs into technical specifications, led Scrum execution, coordinated QA and product teams.

#### Technical Leadership & Architecture

- Optimized build systems: designed Gulp workflow and Jenkins jobs for intelligent parallelization, achieving 70% build time reduction and sustaining peak performance through active task rebalancing.
- Established testing architecture with WebSocket-driven harness for automated testing; prototyped Node.js test orchestration framework using Mocha and Selenium WebDriver.
- Set code quality standards across teams: mandatory code reviews, unit testing, linting, and static analysis integrated into build pipelines.
- Mentored engineering talent through technical deep-dives, code reviews, and hands-on guidance; fostered culture of technical excellence and continuous improvement.
- De-risked architectural decisions by delivering rapid prototypes demonstrating feasibility and technology adoption pathways.

### Digital Chocolate -- Game Engineering Manager -- *(Jun 2009 - Oct 2010, Mexicali, Mexico)*

- Led engineering teams porting mobile games across US carrier devices; established technical quality standards while optimizing resource allocation across multiple concurrent projects.
- Mentored team through hands-on tutorials and code deep-dives on optimization techniques for resource-constrained devices; directly improved team proficiency and project delivery velocity.
- Designed and implemented client module for Qualcomm's Application Value Billing API (C/C++), enabling in-app purchases and establishing reusable code patterns for future game releases.

### LemonQuest -- iPhone Programmer -- *(Apr 2008 - Jun 2009, Salamanca, Spain)*

- Ported PC games from DirectX to iPhone's OpenGL ES, translating desktop mouse/keyboard inputs to touch and accelerometer interactions for mobile user experiences.
- Optimized graphics performance and device interaction patterns across multiple game titles; contributed to studio's transition to emerging iPhone platform.

### Gameloft -- Lead Game Programmer -- *(Aug 2005 - Mar 2008, Mexicali, Mexico)*

- Led first successful iPhone game prototype for the studio (Blockbreaker); demonstrated rapid feature delivery and technical adaptability that earned studio additional iPhone projects.
- Ported multiple game titles to resource-limited, 3D-capable mobile devices using Java/J2ME and OpenGL ES; maintained graphic fidelity and stability within strict memory budgets through careful optimization.
- Led Virgin Mobile catalog team: executed strategic scope reduction, performed asset compression, and applied micro-optimizations to enable shipping games on space-constrained devices while maintaining acceptable performance and quality.

## Professional Development

- Claude Code in Action - Skilljar + Anthropic (2026)
- Introduction to Model Context Protocol - In Progress (2026)

## Education

**Coursework for Master's in Electronics and Telecommunications**

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

**Bachelor of Science in Electronics Systems Engineering**

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)